$$\text{Average Value} = \frac{2Im}{\pi} \text{ or } \frac{2Vm}{\pi}$$

h) Effective Value or RMS Value

The value of an Alternating Current (or) Voltage is equal to the square root of the arithmatic mean of the squares of the instantaneous values taken through one complete cycle.

$$\text{R.M.S. Value} = \frac{Im}{\sqrt{2}} \text{ or } \frac{Em}{\sqrt{2}}$$

A.C ammeters and voltmeters are calibrated to record r.m.s. values.

i) Form Factor

The ratio of r.m.s value to the average value of an alternating quantity is known as form factor

$$\text{Form factor} = \frac{(\text{R.M.S.Value})}{(\text{average value})} = 1.11$$

The form factor is useful in rectifier service because it enables to find the r.m.s.value from average value and vice versa.

j) Power Factor

Cosine value of angle between voltage and current is called as power factor. Power factor is also defined as the ratio of true power to apparent power

Power factor = cos Θ (Θ is angle between voltage and current)

$$\text{Power factor} = \frac{VI\cos\theta}{VI} = \frac{\text{True power}}{\text{Apparent power}}$$

The power factor can never be greater than the value 1. If the powerfactor is 1, it is called as unity power factor. The word lagging or leading with the numerical value of power factor should be noted to signify whether the current lag or leads the voltage.

5.2.1 Phase

Phase of a particular value of an alternating quantity is the fractional part of the time period or cycle through which the quantity has advanced from the selected zero position of reference.

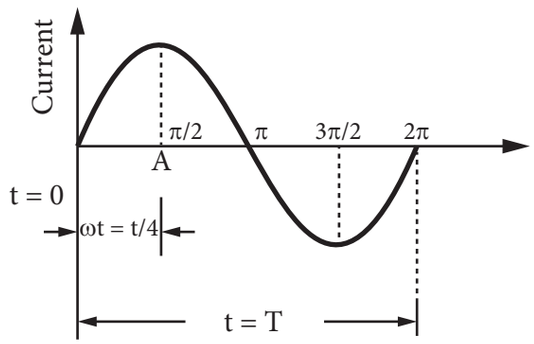


Fig5.5 *Phase*

5.2.2 Phase Difference

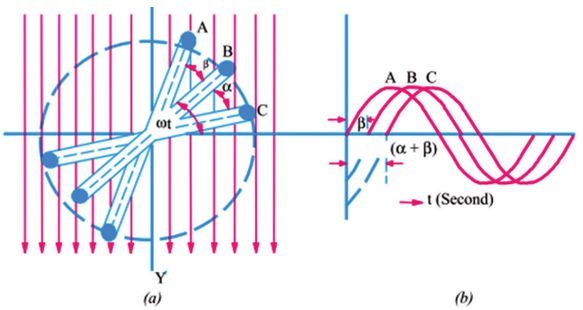


Fig 5.6 *Phase Difference*

When two alternating quantities of the same frequency have different zero points, they are said to have a phase difference.

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